An Exact Finite-p Two-Radius Classification for Horizontal Spherical Means on H-Type Groups

Candidate substantial partial result for arXiv:2109.14963

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Status. Candidate partial result, likely valid, pending specialist review. It settles the source's two-radius question for every finite p and leaves only the nonexceptional $p = \infty$ endpoint.

1 Source problem

E. K. Narayanan, P. K. Sanjay, and K. T. Yasser ask in Remark 4.1 of [1] whether a two-radius theorem holds above their one-radius injectivity threshold $2m/(m-1)$, including $p = \infty$.

> $r\}$. Now the proof can be completed as earlier. We omit the details. This completes the proof of Theorem 1.3.
>
> *Remark* 4.1. The Abel summability result for the spectral decomposition will be true for all $1 < p < \infty$, if we can estimate the operator norm of $\mathcal{A}_k$. It is a natural question whether a two radius theorem is true for functions in $L^p(G)$ for $\frac{2m}{m-1} < p \leq \infty$ and whether our results can be proved for averages over K-orbits where $(G \rtimes K, K)$ is a Gelfand pair as in the case of the Heisenberg group. We hope to return to these questions and some others in the near future.
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> *Remark* 4.2. When $1 \leq p \leq 2$, it is possible to take the Fourier transform in the central variable and prove the injectivity results for the spherical means with weaker conditions of growth on the function. See [DS21]

The complete remark appears on physical page 19 of the arXiv PDF. The part addressed here is:

> "It is a natural question whether a two radius theorem is true for functions in $L^p(G)$ for $\frac{2m}{m-1} < p \leq \infty$."

2 Definitions and spectral input

Let G be an H-type group whose Lie algebra splits orthogonally as $\mathfrak{v} \oplus \mathfrak{z}$, with $\dim \mathfrak{v} = 2n$ and $\dim \mathfrak{z} = m \geq 2$. We write points as $(z, t) \in \mathbb{R}^{2n} \times \mathbb{R}^m$. For $r > 0$, let μ_r be normalized surface measure on $\{(z, 0) : |z| = r\}$, and call $f \mapsto f * \mu_r$ the horizontal spherical mean at radius r.

Let L_k^{n-1} be the generalized Laguerre polynomial. For $k \geq 1$, write its k positive zeros as

$$0 < \zeta_{k,1} < \cdots < \zeta_{k,k},$$

and set

$$\mathcal{E}_n = \left\{ \sqrt{\frac{\zeta_{k,i}}{\zeta_{k,j}}} : k \geq 1,\ 1 \leq i, j \leq k \right\}.$$

This is a countable subset of $(0, \infty)$.

We use two results proved in the source paper [1]. First, its spectral projections $\mathcal{A}_k$ are bounded on $L^p(G)$ for $1 < p < \infty$. Second, for $2 \le p < \infty$,

$$f = \lim_{\rho \uparrow 1} \sum_{k=0}^{\infty} \rho^k \mathcal{A}_k f \quad \text{in } L^p(G). \tag{1}$$

The proof of source Theorem 1.1 also establishes the following support fact before it invokes its low-p Fourier-support theorem.

Lemma 1 (support consequence of the source proof)**.** *Let $2 \le p < \infty$, $f \in L^p(G)$, and $u > 0$. If $f * \mu_u = 0$, then for every $k \ge 0$ and almost every horizontal point z, the distributional Fourier transform in t of $\mathcal{A}_k f(z, \cdot)$ is supported in*

$$Z_{k,u} = \left\{ a \in \mathbb{R}^m : L_k^{n-1}\left(\frac{|a|u^2}{2}\right) = 0 \right\}.$$

The source states this inside the proof for $p \le 2m/(m-1)$, but all steps through the support conclusion use only its L^p boundedness of $\mathcal{A}_k$ and the Abel theorem (1). The upper bound on p first enters in the next sentence, when the authors invoke a theorem forcing an L^p function with Fourier support on a sphere to vanish.

3 Proof Intuition

For one radius, a high-p spectral component may survive because its central Fourier transform can live on a Laguerre zero-sphere. A second radius forces the same component onto a second family of zero-spheres. Unless the ratio of the radii is a ratio of two zeros at the same Laguerre level, the two families do not intersect. The allowed Fourier support is then empty, so no integrability-based support theorem is needed. Abel summability reconstructs f from its now-vanishing spectral components. At the exceptional ratios, the source's explicit spherical eigenfunctions show that this obstruction is real and give common kernel vectors.

4 Exact finite-p classification

Theorem 1. *Let G, m, n, and μ_r be as above, and let $r, s > 0$.*

1. *If $1 \le p \le 2m/(m-1)$, then $f * \mu_r = 0$ already implies $f = 0$ for $f \in L^p(G)$.*
2. *If $2m/(m-1) < p < \infty$, then*

$$f * \mu_r = f * \mu_s = 0 \quad \Longrightarrow \quad f = 0$$

   for every $f \in L^p(G)$ if and only if $r/s \notin \mathcal{E}_n$.

3. *If $p = \infty$ and $r/s \in \mathcal{E}_n$, the pair is not injective. The nonexceptional $p = \infty$ case is not decided here.*

Consequently, the two-radius problem is completely classified on every finite $L^p(G)$.

Proof. Part (1) is source Theorem 1.1 [1]. We prove the new high-p classification.

Assume first that $r/s \notin \mathcal{E}_n$, that $2m/(m-1) < p < \infty$, and that $f * \mu_r = f * \mu_s = 0$. Since $m \geq 2$, the inequality $p > 2m/(m-1)$ implies $p > 2$. Lemma 1, applied at both radii, gives

$$\operatorname{supp} \mathcal{F}_t(\mathcal{A}_k f(z,\cdot)) \subset Z_{k,r} \cap Z_{k,s}$$

for every k and almost every z.

For $k = 0$, $L_0^{n-1} = 1$, hence both zero sets are empty. Let $k \geq 1$. If a belonged to their intersection and $\lambda = |a|$, then $\lambda > 0$ because $L_k^{n-1}(0) \neq 0$. For some i, j we would have

$$\frac{\lambda r^2}{2} = \zeta_{k,i}, \qquad \frac{\lambda s^2}{2} = \zeta_{k,j}.$$

Dividing yields $r/s = \sqrt{\zeta_{k,i}/\zeta_{k,j}} \in \mathcal{E}_n$, a contradiction. Thus every support intersection is empty. A distribution with empty support is zero, so $\mathcal{A}_k f = 0$ for all k. Equation (1) now gives $f = 0$.

Conversely, suppose $r/s \in \mathcal{E}_n$. Choose $k \geq 1$ and roots $\zeta_{k,i}, \zeta_{k,j}$ such that $r/s = \sqrt{\zeta_{k,i}/\zeta_{k,j}}$, and put

$$\lambda = \frac{2\zeta_{k,i}}{r^2} = \frac{2\zeta_{k,j}}{s^2}.$$

The source constructs the nonzero spherical eigenfunction

$$F_{k,\lambda}(z,t) = \frac{J_{m/2-1}(\lambda|t|)}{(\lambda|t|)^{m/2-1}}\, \varphi_k^\lambda(z),$$

up to a harmless normalization, and proves

$$F_{k,\lambda} * \mu_u = c_{k,n} \varphi_k^\lambda(u) F_{k,\lambda}.$$

Our choice of λ makes both $\varphi_k^\lambda(r)$ and $\varphi_k^\lambda(s)$ zero. Hence both means annihilate $F_{k,\lambda}$. The source also proves $F_{k,\lambda} \in L^p(G)$ exactly for $p > 2m/(m-1)$, including $p = \infty$. This proves the converse in (2) and the noninjectivity assertion in (3). $\square$

5 Limitations and novelty status

The theorem does not settle injectivity on $L^\infty(G)$ for $r/s \notin \mathcal{E}_n$. The finite-p proof ends with norm-convergent Abel reconstruction, which the source does not provide at the endpoint. The second source question about averages over general K-orbits is also not addressed.

The local run indexes contained no result for arXiv:2109.14963. Two bounded primary-source searches used the exact title/question, authors, and close variants involving "two radius", "H-type", spherical means, and Laguerre zeros. They found the source and its 2024 publication, and narrower adjacent results, but no statement of the exact all-finite-p classification above. Novelty confidence is therefore moderate rather than definitive.

6 Human review recommendation

Send to a specialist in harmonic analysis on nilpotent Lie groups. The main audit point is Lemma 1: verify directly that the source's support-containment argument remains valid for every finite $p > 2$, and that the source's upper bound on p is used only after that containment has been obtained. A separate citation search under Wiener–Tauberian and spectral synthesis terminology is also recommended.

References

[1] E. K. Narayanan, P. K. Sanjay, and K. T. Yasser, *Injectivity of spherical means on H-type groups*, arXiv:2109.14963; J. Fourier Anal. Appl. **30** (2024), Article 33, doi:10.1007/s00041-024-10089-9.